



GIT VISUALIZATION TOOL

Learn Version Control Through Interactive Visual Simulation

Understanding Git helps you collaborate better and build amazing things.
Let's learn together.

 **START SANDBOX**

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Git Basics

Step 3 of 5: Making Changes

Current Task

You have initialized a repository. Now, try modifying a file and adding it to the staging area.

Tip

Files in the **Working Directory** are untracked or modified. Use `git add` to move them to the **Staging Area**.

1. Initializing a Repository

2. Checking Status

3. Staging Files

WORKING DIRECTORY

Untracked & Modified

3 Files

index.html
MODIFIED

styles.css
UNTRACKED

script.js
UNTRACKED

Use `'git add <file>'` to stage changes

STAGING AREA

Index / Cached

1 File

logo.png
NEW FILE

✓ Commit Staged

STASH



Stash is empty

Use `git stash` to save uncommitted changes temporarily.

REMOTE REPOSITORY

https://github.com/user/repo.git Behind 1 commit



origin/main

Last fetch: 10 mins ago

Push (1)

Fetch

LOCAL REPOSITORY

main branch

HEAD → main



a1b2c3d

2 mins ago

Add login form component

HEAD

main



9f8e7d6

1 hour ago

Initial project structure

origin/main



3a2b1c0

Yesterday

Initial commit

Terminal - bash

3 files changed, 45 insertions(+)

user@devbox:~/project\$ git status

On branch main

Your branch is ahead of 'origin/main' by 1 commit.

user@devbox:~/project \$ |

git status

git add .

git commit

git push

git log --online

 LEARNING CONTEXT

Follow the steps below to master the basic Git workflow.

STEP 3 OF 8



STAGE YOUR CHANGES

You have modified `style.css`. Now, add it to the staging area.

```
git add style.css
```

STEP 4

COMMIT TO HISTORY

 **GIT GLOSSARY**



Working Directory

Your local files

 **index.html**
UNTRACKED

 **style.css**
MODIFIED

 **app.js**
MODIFIED

Staging Area

Files prepared for commit

 **utils.js**
STAGED

 COMMIT STAGED

Local Repository

Your commit history

HEAD: **main**

-  **a1b2c3**
init project
2h ago
-  **d4e5f6**
feat: html structure
1h ago
-  **4f5g6h**
fix: nav
15m ago
main

Remote Repository

Shared repository (origin)



No remote configured. Use 'git remote add origin <url>'

Stash

Temporarily saved changes

CLEAR

Stash is empty

 **STASH CHANGES**

Git Terminal

Help Clear

```
[10:42:01] $ git status
On branch main
Changes not staged for commit:
  modified: style.css
  modified: app.js
```

\$ ~/project git | Type a git command (e.g., git init, git status)...

Execute



Git Add Simulation

Visualize how files move from the Working Directory to the Staging Area.



Reduced Motion

 Working Directory

1 File

Untracked

 Staging Area

0 Files

Ready to commit

UNTRACKED FILES



 script.js

MOD

 styles.css

IGNORED

STAGED CHANGES



Ready to simulate

Click 'Simulate Move' to see the animation.

Reset



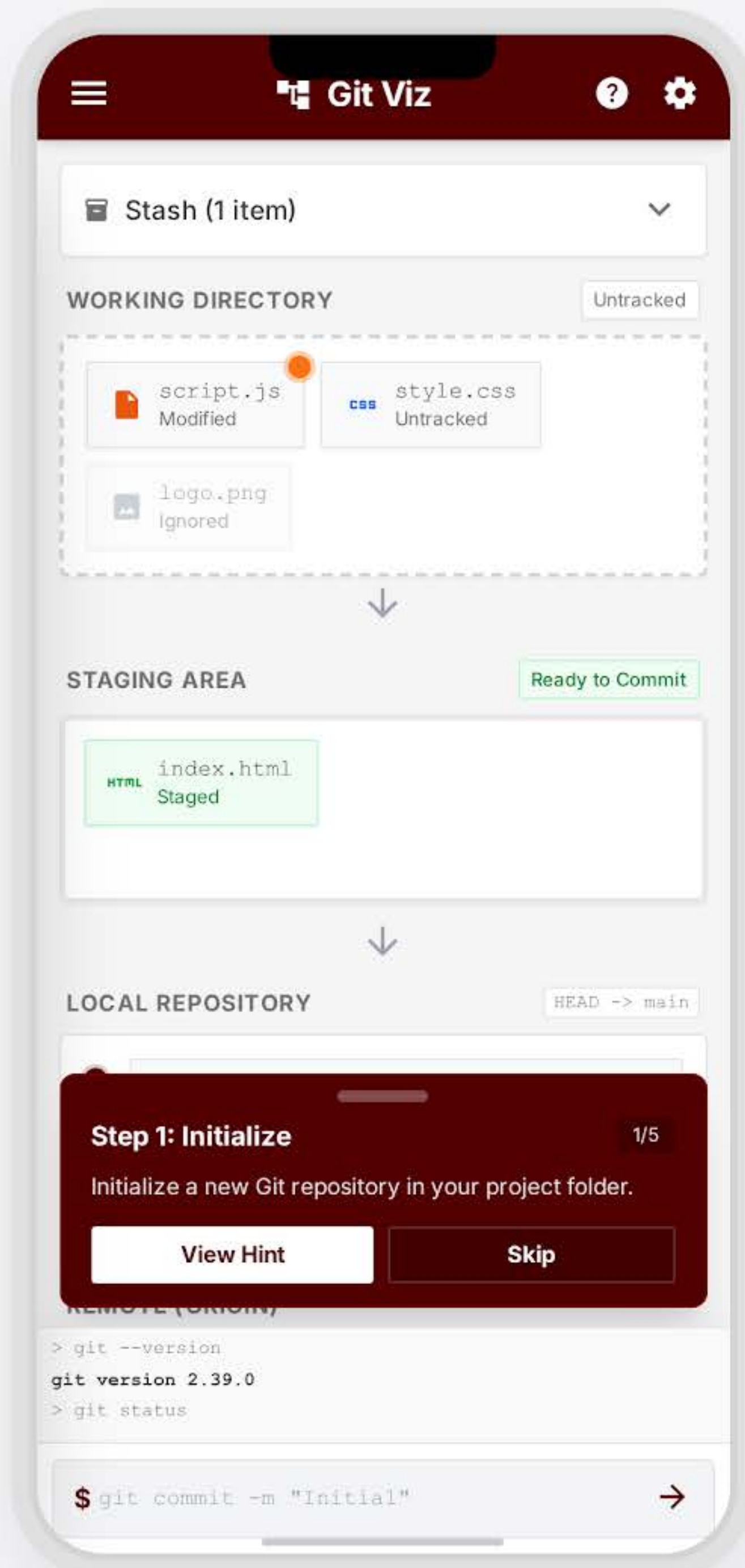
Simulate Move

 System Status

Waiting for input...



Loading Git Visualization Tool...



Git Viz



Stash (1 item)



WORKING DIRECTORY

Untracked



script.js
Modified



style.css
Untracked



logo.png
Ignored



STAGING AREA

Ready to Commit



index.html
Staged



LOCAL REPOSITORY

HEAD -> main

Step 1: Initialize

1/5

Initialize a new Git repository in your project folder.

View Hint

Skip

REMOTE (ORIGIN)

```
> git --version
git version 2.39.0
> git status
```

```
$ git commit -m "Initial"
```

